EXECUTE THE PROGRAM: AUDITING SEARCH AND TOOL USE IN MESH-TO-CAD RECONSTRUCTION

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ABSTRACT

Mesh-to-CAD methods return a program. We ask what an IoU-driven mesh-to-CAD optimizer and its extensions achieve when that program is executed and the solid it builds is judged, instead of the mesh the search reports. We audit CADFit and three search extensions on 60 hard-tier CADBench parts (benchA and benchE) drawn after, and disjoint from, all development data. With beam search, mean mesh IoU is 0.880 but the executed programs score 0.799: 6 programs do not build, and they account for 0.078 of the 0.081 gap. For the rule package and for the full ladder of extensions, executed-program gains are significantly smaller than mesh gains. Whole-part section-extrusion proposals give the largest executed-program increment among the extensions (+0.089 over beam search). We then expose the optimizer's stages (sketch extraction, extrusion and revolve fitting, section extrusion) as tools for six open-weight LLM agents on 20 of these parts. The tools raise every model's executed-program IoU over writing CadQuery alone, by 0.256 to 0.471, with all six Bonferroni-corrected intervals above zero. The best agent, Gemma-4-31B, scores 0.892, against 0.904 for the strongest pipeline arm on the same parts. Against a scripted policy that uses the same tools under the same call budget (**PENDING**), **PENDING**.

1 INTRODUCTION

The product of mesh-to-CAD reconstruction is a program. A designer edits its sketches and features, and a CAD kernel turns it into a solid. Optimization-based reconstructors such as CADFit (Nehme et al., 2026) build that program by proposing extrusions and revolves fitted to slices of the target mesh, composing them with booleans, and keeping the composition with the highest volumetric IoU. In the implementation we audit, the score that steers the search is computed on meshes: candidate solids are tessellated, combined, and compared with the target. The program written out at the end is a separate artifact, and whether it builds and reproduces the scored geometry is not part of the search's number.

Search extensions add compositions, booleans and proposal types, and each is a new way for the reported mesh and the final program to diverge. We therefore measure the divergence arm by arm instead of assuming it is constant. The same question applies to LLM agents for CAD (Mallis et al., 2025; Savrasov et al., 2026), which give a language model geometric tools. CADFather (Savrasov et al., 2026) ablates its tools and leaves a controlled evaluation of the assistant's own choices among them to future work. Separating the two requires a non-LLM policy over the same tools and budget, scored on executed programs.

We study three questions on 60 hard-tier CADBench parts (Doris et al., 2026) drawn after, and disjoint from, every part used during development:

- **Q1** Does mesh IoU track the IoU of the executed program (IoU_{CAD}) across search arms?
- **Q2** Which search extensions improve the executed program, not just the reported mesh?
- **Q3** Do the optimizer's own stages make open-weight LLMs competent CAD agents, and does the LLM add anything over a script that uses the same stages?

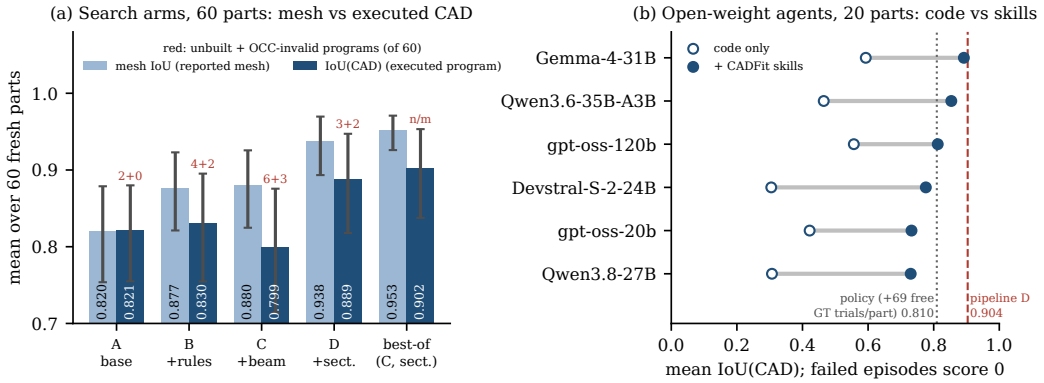


Figure 1: **(a)** Search arms on 60 fresh parts. Light bars score the mesh the search reports; dark bars score the solid built by executing the final program. Error bars: 95% bootstrap intervals of the mean over parts. Red: unbuilt + OCC-invalid programs; n/m = validity not measured for the best-of control. **(b)** Six open-weight agents on 20 of those parts, writing CadQuery alone (open) or with CADFit stages as tools (filled); failed episodes score 0. Gemma-4-31B also shows its three-run range and the code + section-extrusion ablation. Vertical lines: the strongest pipeline arm on the same parts and the two scripted policies over the same tools; the unrestricted policy is labelled with its free ground-truth trials.

Our answers, summarised in Figure 1:

1. **Execution changes the ranking of search arms** (Section 4). The mean gap between mesh IoU and IoU_{CAD} is -0.001 for the base pipeline, 0.081 with beam search and 0.049 with section proposals, and almost all of it comes from programs that do not build. Gains in IoU_{CAD} are significantly smaller than gains in mesh IoU for the rule package and for the full ladder. We also document defects in the search objective and in our own extensions, including an alpha-wrapped objective, cavity filling and self-intersecting lofts, that change individual scores (Section 3).
2. **Whole-part section proposals survive execution** (Section 4). Adding a mid-plane section extrusion per axis to the candidate pool is the largest observed IoU_{CAD} increment among our extensions (+0.089, 95% interval [+0.008, +0.174]). Over the base pipeline, the overall IoU_{CAD} gain of all extensions together is not established, and it is concentrated on benchE.
3. **Optimizer stages are effective tools for open-weight agents** (Section 5). With the stages as tools, all six models improve over writing CadQuery alone, by 0.256 to 0.471 IoU_{CAD}, and every Bonferroni-corrected interval excludes zero. The best agent scores 0.892 against 0.904 for the strongest pipeline arm on the same parts (difference -0.012, interval $[-0.050$, $+0.041]$). Against a scripted policy with the same tools and call budget, **PENDING**.

2 Related work

Mesh- and point-cloud-to-CAD. Learned methods map geometry to construction sequences: DeepCAD (Wu et al., 2021) and Point2Cyl (Uy et al., 2022) predict sketch-and-extrude structure, zone graphs (Xu et al., 2021) search over extrusion sequences, and CAD-Recode (Rukhovich et al., 2025) decodes point clouds into CadQuery code that is executed for evaluation. CADFit (Nehme et al., 2026) optimizes instead: it fits extrusion and revolve candidates to slices of the mesh, composes them greedily to maximize volumetric IoU, then recovers fillets and chamfers, and it reports lower invalid ratios than learned baselines on complex parts. Datasets include ABC (Koch et al., 2019), Fusion 360 Gallery (Willis et al., 2021) and DeepCAD; CADBench (Doris et al., 2026) unifies them with face-count tiers and metrics for fidelity, executability and compactness. Our contribution here is an audit: how an optimizer's search score relates to its replayed final program, and which search extensions survive the replay.

Table 1: Positioning. "Final program executed in eval": the reported score is computed on the solid produced by running the returned program. "Non-LLM policy control": a scripted policy over the same tools and budget.

Work	Input	Tools / proposals	Final program executed in eval	Policy control
CADFit	mesh	built-in pipeline stages	search scored on meshes (our audit of the released code, Section 3)	–
CAD-Recode	point cloud	none (learned decoder)	yes	–
CADReasoner	renders + point cloud	trained iterative editor	yes	–
CAD-Assistant	multimodal queries	FreeCAD API + CAD tools	n/a (task-dependent evaluation)	no
CADFather	mesh	learned + algorithmic tools	yes	no
This work	mesh	CADFit stages as tools	yes (sandboxed replay)	yes

Iterative and agentic CAD generation. Tool-using language models (Yao et al., 2023; Schick et al., 2023) and agent–computer interfaces (Yang et al., 2024) motivate CAD agents that act, observe and revise. CADReasoner (Kabisov et al., 2026) trains a model to edit a runnable CadQuery program from the geometric discrepancy of its previous output, and IterCAD (Hu et al., 2026) trains a multimodal agent against an executable CAD sandbox. CAD-Assistant (Mallis et al., 2025) plans FreeCAD actions with CAD-specific tools such as a sketch parameterizer and a cross-section generator. CADFather (Savrasov et al., 2026) coordinates learned and algorithmic proposal tools with a vision-language assistant, executes the proposed programs and keeps the best valid result. We use frozen open-weight models, take the tools from one optimizer, and add controls aimed at the tool-versus-policy question: the same agents without the tools, one model with a single tool, and two scripted policies that use the tools without an LLM.

3 Setting and measurement

Task and optimizer. Given a watertight triangle mesh, the goal is a CadQuery (CadQuery contributors, 2024) program whose solid matches it. All geometry is normalized so the target is centred and its largest extent spans $[-1, 1]$. CADFit (Nehme et al., 2026) slices the mesh into planar sections, fits lines and arcs to each loop, proposes extrusion candidates from a Chamfer-distance sweep over translations and revolve candidates from an axis analysis, and composes candidates greedily with union and cut. Backward pruning and residual refinement follow, and the final composition is written out as one CadQuery program. In the released code, a composition is scored by concatenating the candidates' meshes and alpha-wrapping (Portaneri et al., 2022) the result before computing IoU.

Defects in objective and evaluation. While extending CADFit on development parts we found defects that change individual scores (Table 2). Two are in the released pipeline: the alpha-wrapped objective fills concave detail (on one smoke-test part the search believed 0.954 where an exact judge measured 0.982), and a final re-measure alpha-wraps multi-body results and can replace a good composition with a worse fallback. Others came from our own extensions: a body-wise union that filled internal cavities, self-intersecting lofts that passed kernel validity while cancelling their own volume, and exact-union scoring, under which the scored mesh is never produced by running the composed program. These are development case studies, not fresh-cohort estimates, and Table 2 states which arms still carry each defect.

Evaluation protocol. The same judge scores every arm and every agent. Mesh IoU compares the method's output mesh with the target: exact volumes from manifold booleans (Lambert & Manifold contributors, 2024) when the output is a closed, consistently wound volume, otherwise a seeded Monte Carlo estimate with generalized-winding-number inside tests (Jacobson et al., 2013). IoU_{CAD} applies the same judge to the solid obtained by running the final program in a fresh, sandboxed process (no network; writes confined to its folder). A program that does not build scores 0. Open-

Table 2: Defects found on development, smoke and earlier test parts (not the fresh cohort), with where each comes from, the case where we observed it, and its status in the fresh arms. Records: `results/defect_cases.json`.

defect	origin	case	before	after	in the fresh arms
Search objective alpha-wraps concatenated candidate meshes before IoU	upstream CADFit	smoke 00053793	search-internal IoU 0.954	strict judge 0.982	fixed from arm B on (exact-union scoring); arm A keeps it
Body-wise union fills internal cavities	our extension (body-wise union)	00344388 (beam); 00014841 (rules)	0.9758; 0.9102	0.9862; 0.9475	fixed in judge for all arms
Scored mesh is a boolean of cached solids; final program never run	our extension (exact-union scoring)	p3test2 beam 00706786; 00409618; 00014841	00706786 scored 0.980 as valid	1/4 solids OCC-invalid; 00409618 2/33 invalid; 00014841 beam program does not build (Null TopoDS_Shape)	evaluation now replays every program
Self-intersecting lofts pass validity checks but cancel volume	our extension (loft/sweep candidates)	one loft candidate; 00399282	greedy believed 0.998	true score 0.776; 15 of 85 loft/sweep candidates rejected on 00399282	loft/sweep not used in any fresh arm
Final re-measure alpha-wraps multi-body results, triggering a worse fallback	upstream CADFit	heatsink 00965166	0.978 composition replaced by 0.878 single extrusion	composition kept	fixed from arm B on (per-body exact union)
Section fallback throws; slices silently dropped	upstream CADFit (with current trimesh)	smoke set	7 slices dropped on 4 smoke parts	0 dropped	fixed in all arms (default on)
Run summary counts crashed parts as successful	upstream CADFit (run summary)	smoke run with a mid-run edit	reported 5/5 successful	3 parts had crashed; only the strict judge catches it	harness trusts only the external judge

Cascade validity of the shape and of every solid is recorded; IoU_{CAD} keeps the measured score of a built but invalid solid, and the validity-gated $\text{IoU}_{\text{CAD}}{}^{\text{valid}}$ is 0 unless the program builds and is valid. Chamfer distance (CD) is the symmetric mean of unsquared nearest-neighbour distances between 20,000 points sampled on each surface (target and output mesh, seed 0) in the normalized frame, averaged over parts with an output, with its denominator stated.

Fresh cohort and statistics. CADBench's benchA and benchE families (Doris et al., 2026) contain ABC-derived parts, the former selected for operation variety and the latter extrusion-based, both stratified by B-rep face count; we use the hard tier. The fresh cohort is a seeded draw of 30 benchA-hard and 30 benchE-hard parts taken after, and disjoint from, all test, development and smoke sets used earlier. All arms run once per part from one code version. Differences are paired by part; intervals are percentile bootstrap over parts (Efron & Tibshirani, 1993) (10,000 resamples, seed 0) and reflect part sampling, not run-to-run randomness. Search-arm comparisons are exploratory with nominal 95% intervals. For the agent study, comparisons across the six models use Bonferroni-corrected 99.17% intervals ($\alpha = 0.05/6$). Win/loss counts use a tie threshold of 10^{-9}.

4 What survives execution: search extensions on a fresh cohort

Arms. All four arms share one code version, seed 0 and one refinement iteration (flags in Appendix A). **A** is the released CADFit with robustness and lossless speed fixes that are on by default in our code (slice fallback, candidate identity in pruning, a refinement guard, caching); it keeps the alpha-wrapped composition objective. **B** adds a rule package: a persistent CadQuery interpreter (byte-identical outputs; interpreter start-up was 89% of a median call), exact-union scoring with per-body re-measure, a cut phase that subtracts candidate solids while IoU improves, event slices between consecutive heights of axis-normal planar faces, extrusion directions from clustered face normals, and a subsampled translation sweep. **C** adds beam search (width 4) over compositions; it also kept intermediate files for a separate study, which does not change the search. **D** adds whole-part section proposals: for each axis, the target's mid-bounding-box cross-section, simplified and extruded through the full extent of the part, enters the candidate pool. This recipe began as a control an external reviewer designed after seeing agent results on test parts; we froze it on development parts before drawing the fresh cohort. As a further control, *best-of* takes, per part, whichever of C's model and the best single section extrusion has the higher IoU against the target mesh.

Mesh IoU overstates the executed program, through programs that do not build (Q1). In the base arm, mesh IoU and IoU_{CAD} agree on average: the mean gap is -0.001 (Table 3). The gap rises to 0.046 with the rule package and 0.081 with beam search, then falls to 0.049 with section proposals.

Table 3: Search arms on 60 fresh parts. IoU_{CAD}: IoU of the solid built by executing the final program (0 if it does not build); $\text{IoU}_{\text{CAD}}^{\text{valid}}$: additionally 0 unless the shape is OpenCascade-valid. CD over parts with an output (n). Unbuilt + invalid: programs that do not build + built but invalid. n/m: not measured.

	mesh IoU	IoU_{CAD}	$\text{IoU}_{\text{CAD}}^{\text{valid}}$	gap	CD (n)	unbuilt + inv.	IoU_{CAD} benchA	IoU_{CAD} benchE
A base	0.820	0.821	0.821	−0.001	0.0155 (58)	2 + 0	0.866	0.777
B + rules	0.877	0.830	0.799	0.046	0.0130 (59)	4 + 2	0.837	0.824
C + beam	0.880	0.799	0.753	0.081	0.0129 (59)	6 + 3	0.775	0.824
D + sections	0.938	0.889	0.856	0.049	0.0105 (59)	3 + 2	0.859	0.918
best-of (C, sections)	0.953	0.902	n/m			n/m		
sections only	0.842							

paired difference	Δ mesh IoU	Δ IoU_{CAD}	Δ $\text{IoU}_{\text{CAD}}^{\text{valid}}$	ΔIoU_{CAD} − Δmesh
B − A (rules)	+0.056 [+0.027, +0.098]	+0.009 [−0.058, +0.070]	−0.022 [−0.100, +0.050]	−0.047 [−0.105, −0.003]
C − B (beam)	+0.003 [+0.0004473, +0.007]	−0.031 [−0.094, +0.029]	−0.047 [−0.122, +0.016]	−0.034 [−0.097, +0.026]
D − C (sections)	+0.058 [+0.026, +0.101]	+0.089 [+0.008, +0.174]	+0.103 [+0.029, +0.182]	+0.032 [−0.044, +0.109]
D − A (all)	+0.117 [+0.070, +0.175]	+0.067 [−0.012, +0.144]	+0.035 [−0.057, +0.121]	−0.050 [−0.109, −0.003]

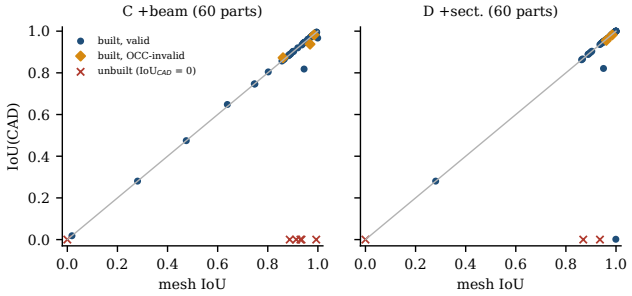


Figure 2: Per-part mesh IoU against IoU_{CAD} for arms C and D (60 parts). Red crosses: programs that do not build (IoU_{CAD} = 0); orange diamonds: built but OpenCascade-invalid. The diagonal marks equal mesh IoU and IoU_{CAD}.

In arm C, 6 programs do not build and 3 build but fail OpenCascade validity. Averaged over all 60 parts, the unbuilt programs contribute 0.078 of the 0.081 gap and built programs 0.003. For built programs the two scores usually agree, though not always: the largest built-program difference is 0.127 in arm C, and in arm D one valid program scores 0.002 against a mesh IoU of 0.999 (Figure 2). Kernel validity alone therefore does not certify the reconstruction.

Gains shrink when programs are executed (Q2). Each extension raises mean mesh IoU, and every mesh-IoU interval in Table 3 excludes zero. The last column tests whether the executed-program gain is smaller than the mesh gain on the same parts. It is for the rule package (−0.047, [−0.105, −0.003]) and for the full ladder D − A (−0.050, [−0.109, −0.003]); for beam search alone the test is inconclusive ([−0.097, +0.026]). Win/loss counts show how a few failures dominate a mean: under $\text{IoU}_{\text{CAD}}^{\text{valid}}$, the rule package improves 39 parts and worsens 7, yet its mean change is −0.022, because 5 of the worsened parts lose their whole score.

Whole-part section proposals survive execution (Q2). Section proposals are the largest observed IoU_{CAD} increment among the extensions: D − C is +0.089 [+0.008, +0.174], with similar point estimates on benchA (+0.085) and benchE (+0.094), and D has the highest IoU_{CAD} of the four arms. A single best-axis section extrusion with no search reaches a mean mesh IoU of 0.842, above the base arm's 0.820. Folding the proposals into the search is not distinguishable from picking the better of C and the best section per part (D − best-of on IoU_{CAD}: −0.014 [−0.080, +0.053]). Relative to the base arm, the gains and losses of the whole ladder do not net out to an established IoU_{CAD} improvement: D − A is +0.067 [−0.012, +0.144]. That net change is concentrated on benchE (+0.141 [+0.025, +0.261]) rather than benchA (−0.006 [−0.112, +0.073]). The per-split estimates locate the gain without testing an interaction, and because each arm adds its rules as a package, the design separates packages, not individual rules.

5 OPTIMIZER STAGES AS TOOLS FOR OPEN-WEIGHT AGENTS

Having identified which geometric proposals survive execution, we test what LLM agents gain from access to the optimizer's operations, and whether a script with the same operations does as well.

Harness. An agent receives no image or mesh. It learns about the target only through tools, over an OpenAI-compatible tool-calling loop served by vLLM (Kwon et al., 2023) on one GPU per model. Every toolset includes `describe_target` (size, volume, flat faces), `section` (exact cross-section polygons of the target and the current model on an axis-aligned plane), `evaluate` (IoU against the target and where material is missing or extra), `add_code` (union or cut of a solid built by agent-written CadQuery), `remove_step`, `list_steps` and `finish`. The **code** toolset is exactly this. The **skills** toolset adds five CADFit stages: `detect_directions`; `extract_sketch`, which slices the target at a plane, nests loops into profiles and fits lines and arcs; `fit_extrude` (translation sweep, then extents); `fit_revolve` (axis analysis); and `section_extrude` (the whole-part proposal of Section 4). Each stage's solids become candidates, scored against the target when created, which the agent can list, inspect and add. The **code+section** toolset adds only `section_extrude` and the candidate tools; we run it for Gemma-4-31B only. The system prompts differ only in the tool paragraph and one strategy phrase. Episodes start from an empty model, allow 40 tool calls, 30 minutes and 16,384 generated tokens per turn, and return the best model seen, as judged by IoU against the target; a failed episode scores 0. Target-score feedback is part of the task, since the target mesh is the method's input. Agent-written code runs in a sandbox. Three implementation details make the skills differ from the frozen pipeline arms: even-odd loop nesting (the pipeline treated a material island inside a hole as a second hole), a 10^{-6} minimum profile area (the pipeline's 10^{-3} dropped all 67 profiles of one slice), and a fix to how revolve profiles are split at the axis (a stepped shaft went from 0.408 to 0.994 IoU).

Models and parts. Six open-weight models, each on one GPU, with model-card sampling and reasoning enabled where the model supports it: Gemma-4-31B (Google DeepMind, 2026), Qwen3.6-35B-A3B and Qwen3.8-27B (Yang et al., 2025), gpt-oss-120b and gpt-oss-20b (OpenAI et al., 2025), and Devstral-Small-2-24B (Mistral AI, 2025). We use 20 parts (12 benchA, 8 benchE) drawn by seed from the fresh cohort before any episode ran, with one episode per model, toolset and part.

Scripted policies. The *unrestricted* policy calls `section_extrude` on each axis, extracts mid-plane sketches along the detected directions, fits the three largest profiles per plane, and assembles greedily by best improvement. Its assembly scores every candidate union and cut against the target without spending calls: 69 free trials per part on average. The *budget-matched* policy charges each such trial as one of its 40 calls, plus one more call to add an accepted candidate; it keeps the first improving trial (candidates in IoU order, adds before cuts) and reserves 20 calls for assembly. The current model's own score, recomputed before each trial, is not charged. It used **PENDING** calls per part, **PENDING** of them trials. For comparison, an agent adding a candidate spends one call and sees the new IoU, and undoing a bad addition costs a second call; skills candidates arrive already scored against the target.

The stages make every model a better CAD agent (Q3). With the stages as tools, all six models improve over writing CadQuery alone (Table 4), by 0.256 to 0.471 IoU_{CAD}, and every Bonferroni-corrected interval excludes zero. The largest gains go to the weakest code-writers: Devstral rises from 0.305 to 0.776 and Qwen3.8 from 0.307 to 0.730. Validity gating lowers a skills score by at most 0.044. Gemma-4-31B with skills reaches 0.892, an observed -0.012 relative to pipeline D on the same parts ([-0.050, +0.041]). Its three runs over the 20 parts score **PENDING**, **PENDING** and **PENDING** (mean **PENDING**, range **PENDING**–**PENDING**); the first is the declared primary result, and the range describes three runs, not an interval.

Ablation: code plus section extrusion only (Gemma). With only `section_extrude` beyond the code tools, Gemma scores **PENDING**. Relative to code alone the difference is **PENDING PENDING**, and full skills minus code+section is **PENDING PENDING** (nominal 95% intervals).

Agents versus scripted policies. The last two columns of Table 4 compare each model's skills episodes with the two scripted policies. Against the unrestricted policy (0.810), every Bonfer-

Table 4: Agents on 20 fresh parts, mean IoU_{CAD} (failed episodes score 0). Differences are paired over parts, with Bonferroni-corrected 99.17% bootstrap intervals ($\alpha = 0.05/6$) below. References on the same parts: pipeline D 0.904; unrestricted scripted policy 0.810 (69 free target-score trials per part); budget-matched policy **PENDING**. Win/loss counts and validity-gated scores: Table 5.

model	code	skills	skills − code	skills − pipeline D	skills − unrestricted policy	skills − matched policy
Gemma-4-31B	0.593	0.892	+0.299 [+0.138, +0.481]	−0.012 [−0.050, +0.041]	+0.083 [−0.0001087, +0.234]	**PENDING** **PENDING**
Qwen3.6-35B-A3B	0.465	0.854	+0.389 [+0.206, +0.567]	−0.050 [−0.122, +0.026]	+0.044 [−0.034, +0.204]	**PENDING** **PENDING**
gpt-oss-120b	0.557	0.812	+0.256 [+0.112, +0.417]	−0.091 [−0.221, +0.010]	+0.003 [−0.137, +0.184]	**PENDING** **PENDING**
Devstral-S-2-24B	0.305	0.776	+0.471 [+0.319, +0.613]	−0.128 [−0.260, −0.001]	−0.033 [−0.167, +0.150]	**PENDING** **PENDING**
gpt-oss-20b	0.422	0.732	+0.311 [+0.104, +0.519]	−0.171 [−0.340, −0.047]	−0.077 [−0.249, +0.127]	**PENDING** **PENDING**
Qwen3.8-27B	0.307	0.730	+0.423 [+0.178, +0.663]	−0.174 [−0.369, −0.031]	−0.080 [−0.305, +0.146]	**PENDING** **PENDING**

roni interval includes zero; Gemma's comes closest (+0.083, [−0.0001087, +0.234]). Against the budget-matched policy (**PENDING**), **PENDING**.

How the agents use the stages. Averaged per model over its episodes, agents call `extract_sketch` 1.8–4.9 times and `fit_extrude` 1.6–6.8 times, but `fit_revolve` at most 0.3 times (Appendix D). Qwen3.8's most common failure is not geometric. At 16,384 tokens per turn, 16 of 20 Qwen3.8 code episodes terminated on a turn without a tool call. In all 16 of them, the preceding reply had hit the token limit (the harness then asks the model to think briefly and call a tool), and the final reply was 32.5–39.4 thousand characters of reasoning with no content (Appendix E).

6 Discussion and conclusion

Executing the returned program changes what a mesh-to-CAD search appears to achieve. Every extension we measured raises reported mesh IoU, but the extensions also produce programs that do not build, and for the rule package and the full ladder the executed-program gain is significantly smaller than the mesh gain. Whole-part section proposals hold up best under execution. Offered as tools, the optimizer's stages raise every one of six open-weight models well above what it achieves by writing CadQuery alone, and the best agent scores 0.892 against 0.904 for the strongest pipeline arm on the same parts. A scripted policy with the same tools and call budget scores **PENDING**; **PENDING**. CAD-agent evaluations that report only the agent cannot make this distinction, and scoring the executed program is what makes the comparison meaningful.

Limitations. (i) Each search arm and each agent condition ran once per part, except three Gemma runs; our intervals capture part sampling, not run-to-run variation. (ii) The agent study uses 20 parts, so its intervals are wide, and the policy comparisons in particular have little power. (iii) The cohort is hard-tier benchA/benchE. Agents rarely called `fit_revolve`, so revolve-heavy reconstruction is not exercised. (iv) We audit one optimizer family; the mesh–program gap is a property of these searches, and other reconstructors need their own replay audit. (v) The agent's revolve tool includes a profile-split fix that the frozen pipeline arms lack, and the section-proposal recipe originated from a control designed after seeing agent results on test parts, although it was frozen before the fresh cohort was drawn. (vi) All conditions select answers with IoU against the target mesh; this is the task's input, but the results do not transfer to settings without that feedback.

REPRODUCIBILITY STATEMENT

Numbers, tables and figures are generated from result files. `bench/paper_stats.py` computes `results/*.json` from the per-part scores (`bench/runs/fresh_scores.json`, `fresh_section_control.json`, `agentsk_*scores.json`); `bench/paper_numbers.py` writes the macros in `paper/numbers.tex`, `bench/paper_tables.py` the generated tables, and `paper/figures/make_figures.py` the figures. Development case values are recorded with their sources in `results/dev_facts.json` and `results/defect_cases.json`; configuration constants (budgets, thresholds) are set in the scripts that use them. Per-part scores come from `bench/agentsk_score.py` and `bench/score.py -cad`. Model identifiers, serving flags and sampling settings are in `bench/agentsk_models.py`; arm flags are in `bench/run_parallel_arms.sh` (Appendix A); the cohort manifest is `bench/cadbench/index.json` with the fresh file list, and the agent sample is `study_parts` with seed 1. The pipeline arms ran from per-arm code snapshots taken at launch; the analysis and agent code correspond to repository tag `paper-v1`. Each design decision of the agent study that the task left open is recorded, with its rationale, in the decision ledger `implement-stage/ASSUMPTIONS.md`.

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A ARM CONFIGURATIONS

All arms: `-max-iterations 1 -seed 0`, one part per process, frozen code snapshot per arm (`bench/run_parallel_arms.sh`).

arm	flags added to the defaults
A base	none (defaults: slice fallback fix, candidate identity in pruning, IoU validity check, refinement guard, volume/solid caching, persistent pool, lossless CD sweep, at most 200 loops per slice)
B rules	`-persistent-cq -exact-union -cut-phase -event-slices -detect-directions -translation-max-points 400`
C beam	B + `-beam 4 -keep-history` (`-keep-history` only skips deleting intermediate files after the run)
D sections	B + `-beam 4 -section-extrude`

B AGENT HARNESS DETAILS

Sandbox. Agent-written code and the replay of every final program run in a sandbox with no network and writes confined to the episode folder: bubblewrap where available; otherwise an unprivileged user namespace in which every mount is remounted read-only and verified (the sandbox refuses to run if any mount stays writable), temporary directories are replaced by empty file systems, the episode folder is bound back read-write, and all capabilities are dropped with no-new-privileges set.

Context. The last six tool results are kept in full; older results, older assistant text and older code arguments are truncated to 300 characters in what is sent back. Tool results are capped at 8,000 characters while remaining valid JSON. On a context-length error the kept window halves.

Models.

model	temperature	request extras and vLLM parsers
Gemma-4-31B (`google/gemma-4-31B-it`)	1.0	`{"top_p": 0.95, "top_k": 64, "chat_template_kwargs": {"enable_thinking": true}}; -reasoning-parser gemma4 -tool-call-parser gemma4`
Qwen3.6-35B-A3B (`Qwen/Qwen3.6-35B-A3B-FP8`)	1.0	`{"top_p": 0.95, "top_k": 20, "min_p": 0.0, "presence_penalty": 0.0, "chat_template_kwargs": {"enable_thinking": true}}; -reasoning-parser qwen3 -tool-call-parser qwen3_coder -language-model-only`
gpt-oss-120b (`openai/gpt-oss-120b`)	1.0	`{"top_p": 1.0}; -reasoning-parser openai_gptoss -tool-call-parser openai`
Devstral-S-2-24B (`mistralai/Devstral-Small-2-24B-Instruct-2512`)	0.15	`{}; -tool-call-parser mistral -tokenizer-mode mistral -config-format mistral -load-format mistral`
gpt-oss-20b (`openai/gpt-oss-20b`)	1.0	`{"top_p": 1.0}; -reasoning-parser openai_gptoss -tool-call-parser openai`
Qwen3.8-27B (`Qwen/Qwen3.8-27B-FP8`)	1.0	`{"top_p": 0.95, "top_k": 20, "min_p": 0.0, "presence_penalty": 0.0, "chat_template_kwargs": {"enable_thinking": true}}; -reasoning-parser qwen3 -tool-call-parser qwen3_xml -language-model-only`

Prompts. The three system prompts differ only in the tool paragraph and the strategy phrase.

System prompt, `code` toolset.

```
You are a CAD engineer. Your job: build a solid CAD model that matches a target 3D part as
    closely as possible, measured by volumetric IoU (intersection over union) against the
    target.

You cannot see the target. You learn about it only through tools:
- describe_target: size, volume, and the flat faces (directions and plane positions).
- section: exact cross-section polygons of the target (and of your model) in any
    axis-aligned plane.
- add_code: append a solid built by your CadQuery code to the model as a union ('add') or a
    difference ('cut').
- evaluate: current IoU and where material is missing or extra.
- remove_step, list_steps, finish.

Coordinates: everything is in the normalized frame (the target is centered and scaled so
    its largest extent spans [-1, 1]). Your CadQuery code (cq and math are imported) must
    assign the final solid to `result`, e.g.
    result = cq.Workplane('XY').box(1.2, 0.8, 0.1).translate((0, 0, 0.35))
    result = cq.Workplane('XY', origin=(0, 0, -0.2)).circle(0.3).extrude(0.4)
    result =
    cq.Workplane('XZ').polyline([(0,0),(0.5,0),(0.5,0.2),(0,0.2)]).close().revolve(360,
    (0,0,0), (0,1,0))

Good strategy: describe the target, read sections to get exact profile shapes, build the
    main body first, then add features and cut holes/pockets, checking evaluate after
    changes and removing steps that lower IoU. The best model seen at any point is kept, so
    experimenting is safe. Work efficiently: you have a limited number of tool calls. Call
    finish when you cannot improve further.
```

System prompt, `codesec` toolset.

```
You are a CAD engineer. Your job: build a solid CAD model that matches a target 3D part as
    closely as possible, measured by volumetric IoU (intersection over union) against the
    target.

You cannot see the target. You learn about it only through tools:
- describe_target: size, volume, and the flat faces (directions and plane positions).
- section: exact cross-section polygons of the target (and of your model) in any
    axis-aligned plane.
- section_extrude: a CADFit stage that extrudes the target's whole cross-section at a plane
    through the full extent of the part. Each solid it makes becomes a scored candidate:
    list_candidates / show_candidate show them; precision near 1 means the candidate lies
    inside the target.
- add_candidate / add_code: append a solid to the model as a union ('add') or a difference
    ('cut').
- evaluate: current IoU and where material is missing or extra.
- remove_step, list_steps, finish.

Coordinates: everything is in the normalized frame (the target is centered and scaled so
    its largest extent spans [-1, 1]). Your CadQuery code (cq and math are imported) must
    assign the final solid to `result`, e.g.
    result = cq.Workplane('XY').box(1.2, 0.8, 0.1).translate((0, 0, 0.35))
    result = cq.Workplane('XY', origin=(0, 0, -0.2)).circle(0.3).extrude(0.4)
```

```
    result =
    cq.Workplane('XZ').polyline([(0,0),(0.5,0),(0.5,0.2),(0,0.2)]).close().revolve(360,
    (0,0,0), (0,1,0))

Good strategy: describe the target, read sections to get exact profile shapes and extrude
    whole cross-sections, build the main body first, then add features and cut
    holes/pockets, checking evaluate after changes and removing steps that lower IoU. The
    best model seen at any point is kept, so experimenting is safe. Work efficiently: you
    have a limited number of tool calls. Call finish when you cannot improve further.
```

System prompt, `skills` toolset.

```
You are a CAD engineer. Your job: build a solid CAD model that matches a target 3D part as
    closely as possible, measured by volumetric IoU (intersection over union) against the
    target.

You cannot see the target. You learn about it only through tools:
- describe_target: size, volume, and the flat faces (directions and plane positions).
- section: exact cross-section polygons of the target (and of your model) in any
    axis-aligned plane.
- CADFit skills, the stages of a CAD reconstruction pipeline, which you run yourself:
    detect_directions (likely extrusion directions), extract_sketch (slice the target at a
    plane into sketch profiles fitted with lines and arcs), fit_extrude / fit_revolve (turn
    a sketch into extrusion / revolve solids that match the target), section_extrude
    (extrude a whole cross-section through the part). Each solid they make becomes a scored
    candidate: list_candidates / show_candidate show them; precision near 1 means the
    candidate lies inside the target.
- add_candidate / add_code: append a solid to the model as a union ('add') or a difference
    ('cut').
- evaluate: current IoU and where material is missing or extra.
- remove_step, list_steps, finish.

Coordinates: everything is in the normalized frame (the target is centered and scaled so
    its largest extent spans [-1, 1]). Your CadQuery code (cq and math are imported) must
    assign the final solid to `result`, e.g.
    result = cq.Workplane('XY').box(1.2, 0.8, 0.1).translate((0, 0, 0.35))
    result = cq.Workplane('XY', origin=(0, 0, -0.2)).circle(0.3).extrude(0.4)
    result =
    cq.Workplane('XZ').polyline([(0,0),(0.5,0),(0.5,0.2),(0,0.2)]).close().revolve(360,
    (0,0,0), (0,1,0))

Good strategy: describe the target, extract sketches on its main planes and fit them into
    candidates, build the main body first, then add features and cut holes/pockets,
    checking evaluate after changes and removing steps that lower IoU. The best model seen
    at any point is kept, so experimenting is safe. Work efficiently: you have a limited
    number of tool calls. Call finish when you cannot improve further.
```

C AGENT STUDY: VALIDITY-GATED SCORES AND WIN/LOSS COUNTS

Table 5: Validity-gated $\text{IoU}_{\text{CAD}}^{\text{valid}}$ (0 unless the program builds and is OpenCascade-valid) and win–loss counts of the skills condition against code, pipeline D and the unrestricted policy (tie threshold 10^{-9}). Pipeline D $\text{IoU}_{\text{CAD}}^{\text{valid}}$ on these parts: 0.904; unrestricted policy 0.810; budget-matched policy **PENDING**.

model	$\text{IoU}_{\text{CAD}}^{\text{valid}}$ code	$\text{IoU}_{\text{CAD}}^{\text{valid}}$ skills	W–L vs code	W–L vs pipeline D	W–L vs unrestricted
Gemma-4-31B	0.580	0.892	17–3	6–10	10–4
Qwen3.6-35B-A3B	0.465	0.854	17–3	5–11	5–7
gpt-oss-120b	0.557	0.769	18–2	4–13	6–7
Devstral-S-2-24B	0.296	0.737	19–1	3–16	4–14
gpt-oss-20b	0.422	0.732	15–5	2–14	2–12
Qwen3.8-27B	0.307	0.730	15–3	3–15	8–10

D TOOL USE

Tool calls per episode in the skills toolset, over the 20 episodes of each model (`results/tool_usage.json`).

model	calls	extract_sketch	fit_extrude	fit_revolve	section_extrude	detect_directions	section	add_candidate	add_code	evaluate	remove_step	errors
Gemma-4-31B	27.9	3.1	2.7	0.1	1.1	0.4	6.0	2.9	3.5	1.9	2.6	17/559
Qwen3.6-35B-A3B	37.2	4.9	2.2	0.0	1.6	1.0	12.3	2.5	3.4	2.2	2.5	47/744
gpt-oss-120b	38.2	4.8	3.6	0.1	1.6	1.0	6.6	4.0	3.6	3.0	3.4	22/764
Devstral-S-2-24B	37.2	3.8	6.8	0.3	1.2	0.9	2.7	7.2	3.4	4.1	3.9	12/745
gpt-oss-20b	25.1	2.1	1.8	0.1	1.6	0.9	6.2	2.6	2.2	2.6	1.2	44/503
Qwen3.8-27B	22.2	1.8	1.6	0.0	0.3	1.0	7.9	1.3	3.1	1.3	1.8	27/445

E QWEN3.8 TOKEN-LIMIT FAILURES AND A POST-HOC SENSITIVITY RUN

The harness sends "Your reply hit the token limit before you called a tool. Think briefly, then call a tool now." only when a reply ends because it reached the 16,384-token limit; after two consecutive nudges without a tool call the episode stops. In all 16 Qwen3.8 code episodes that stopped this way, the last nudge was the token-limit message, and the final reply was 32.5–39.4 thousand characters of reasoning with no content (`results/qwen38_traces.json`). After seeing these outcomes we reran Qwen3.8 on the same parts with 32,768 tokens per turn. It scored 0.783 with skills and 0.541 with code (versus 0.730 and 0.307 at 16,384). Because this run was decided after seeing results, it is excluded from Table 4; most of its episodes ended at the 30-minute limit instead.

F REFINING PIPELINE OUTPUT WITH AN LLM AGENT

In an earlier study, Qwen3.8-27B (INT4) refined arm C's output on the 19 fresh parts whose C mesh IoU was below 0.90 (gate fixed before running), with access to C's candidate pool, 40 calls and 20 minutes, in two runs. Mean mesh IoU: pipeline 0.751; agent 0.797 and 0.807 (wins–losses against the pipeline at a ± 0.01 threshold: 2–3 and 4–3); a no-op agent that only re-exports the pipeline model 0.749; an equal-time scripted search over the same pool 0.756 (8–2). The mean gain comes from a few parts and does not repeat consistently. Per-part records were not retained, so no intervals or executed-program scores are available for this study.

G PER-PART RESULTS

part	split	A mesh	A CAD	B mesh	B CAD	C mesh	C CAD	D mesh	D CAD
00038011	A	0.844	0.844	0.939	0.937	0.940	0.940	0.940	0.940
00046097	A	0.896	0.915	0.939	0.940	0.942	0.943	0.942	0.943
00053172	E	0.994	0.994	0.994	0.994	0.994	0.994	0.998	0.998
00090768	A	0.962	0.962	0.971	0.971	0.999	0.967	0.999	0.002
00095369	A	0.882	0.882	0.893	0.893	0.893	0.893	0.948	0.948
00101419	A	0.965	0.965	0.978	0.978	0.978	0.978	1.000	1.000
00140273	E	0.884	0.884	0.884	0.884	0.884	0.884	0.894	0.894
00153120	E	0.771	0.771	0.857	0.857	0.857	0.857	0.996	0.996
00154184	E	0.000	–†	0.980	0.981	0.984	0.984	0.984	0.984
00161800	A	0.985	0.985	0.985	0.985	0.985	0.985	0.985	0.985
00181393	A	0.863	0.863	0.863	0.863	0.863	0.863	0.863	0.863
00208290	E	0.989	0.989	0.989	0.989	0.989	0.989	0.992	0.992
00214289	A	0.997	0.997	0.989	0.989	0.989	0.989	0.989	0.989
00232213	E	0.837	0.837	0.869	0.000†	0.861	0.873*	0.869	0.000†
00246395	E	0.959	0.959	0.959	0.959	0.959	0.959	0.997	0.997
00247424	E	0.000	–†	0.000	–†	0.000	–†	0.000	–†
00249450	E	0.926	0.926	0.960	0.960	0.960	0.960	0.995	0.995
00256926	E	0.814	0.814	0.892	0.892	0.899	0.902	1.000	1.000
00266468	E	0.723	0.724	0.746	0.746	0.749	0.746	1.000	1.000
00273535	E	0.912	0.912	0.995	0.995	0.995	0.995	1.000	1.000
00299355	E	0.975	0.976	0.975	0.976	0.975	0.976	0.990	0.990
00305871	E	0.960	0.960	0.961	0.961	0.961	0.961	1.000	1.000
00312587	E	0.973	0.973	0.975	0.974	0.975	0.974	1.000	1.000
00317970	E	0.182	0.182	0.733	0.743	0.746	0.747	0.993	0.993
00354176	A	0.896	0.896	0.903	0.904	0.903	0.904	0.903	0.904
00365747	A	0.848	0.848	0.880	0.000†	0.888	0.000†	0.961	0.956*
00382121	A	0.920	0.920	0.933	0.933	0.936	0.934	0.936	0.936
00492401	A	0.975	0.986	0.986	0.986	0.987	0.988	0.987	0.987

part	split	A mesh	A CAD	B mesh	B CAD	C mesh	C CAD	D mesh	D CAD
00528181	A	0.860	0.876	0.915	0.916*	0.882	0.882	0.960	0.958
00562677	A	0.867	0.867	0.889	0.889	0.889	0.889	0.889	0.889
00589302	A	0.921	0.921	0.966	0.966	0.969	0.937*	0.949	0.949
00630878	A	0.848	0.849	0.942	0.944	0.944	0.944	0.944	0.944
00634093	A	0.931	0.931	0.941	0.942	0.946	0.946	0.946	0.946
00642725	A	0.908	0.908	0.931	0.000†	0.930	0.000†	0.936	0.000†
00657371	E	0.901	0.902	0.901	0.902	0.902	0.903	0.987	0.987
00678171	E	0.935	0.935	0.945	0.945	0.945	0.945	0.949	0.949
00700032	A	0.855	0.855	0.885	0.881	0.886	0.886	0.955	0.955
00709520	A	0.888	0.889	0.888	0.889	0.888	0.889	0.888	0.889
00729452	A	0.051	0.053	0.280	0.281	0.280	0.281	0.280	0.281
00730725	E	0.018	0.018	0.018	0.018	0.018	0.018	0.949	0.949
00771160	E	0.890	0.900	0.974	0.974	0.985	0.982*	0.985	0.982*
00771600	E	0.919	0.919	0.919	0.919	0.920	0.920	0.999	0.999
00780711	E	0.911	0.911	0.973	0.973	0.983	0.983	0.983	0.983
00791454	E	0.970	0.970	0.970	0.970	0.970	0.970	0.990	0.990
00795057	E	0.949	0.949	0.969	0.969	0.980	0.980	0.997	0.997
00814797	E	0.933	0.933	0.970	0.970	0.970	0.970	0.999	0.999
00834538	A	0.920	0.920	0.948	0.948	0.954	0.949	0.968	0.968
00846352	A	0.859	0.859	0.859	0.859	0.867	0.867	0.867	0.867
00862084	A	0.593	0.593	0.553	0.557	0.639	0.648	0.999	0.999
00862085	E	0.422	0.422	0.475	0.475	0.475	0.475	0.999	0.999
00862953	E	0.796	0.796	0.802	0.804	0.802	0.804	0.980	0.980
00883892	E	0.792	0.792	0.905	0.906	0.913	0.000†	0.900	0.900
00892314	A	0.916	0.916	0.933	0.940*	0.935	0.000†	0.959	0.959
00904252	A	0.977	0.977	0.977	0.977	0.977	0.977	0.977	0.977
00941601	A	0.923	0.923	0.943	0.943	0.947	0.947	0.949	0.949
00953658	E	0.973	0.973	0.976	0.976	0.978	0.978	0.982	0.982
00966049	A	0.815	0.815	0.945	0.818	0.945	0.818	0.950	0.821
00966713	E	0.994	0.994	0.994	0.994	0.994	0.994	0.998	0.998
00983172	A	0.877	0.877	0.993	0.985	0.993	0.000†	0.984	0.984
00992810	A	0.879	0.879	0.995	0.995	0.995	0.995	0.997	0.997

† program does not build; * builds but fails OpenCascade validity.

Table 7: Per-part $\mathrm{IoU}_{\mathrm{CAD}}$ of the agent study (failures 0).

part	Gemma-4-31B		Qwen3.6-35B-A3B		gpt-oss-120b		Devstral-S-2-24B		gpt-oss-20b		Qwen3.8-27B		policy	pipe D
	code	skills	code	skills	code	skills	code	skills	code	skills	code	skills		
00046097	0.389	0.832	0.000	0.739	0.702	0.770	0.274	0.765	0.844	0.739	0.000	0.785	0.776	0.943
00095369	0.955	0.951	0.945	0.872	0.866	0.953	0.086	0.838	0.873	0.872	0.945	0.986	0.899	0.948
00140273	0.251	0.894	0.439	0.884	0.446	0.894	0.308	0.894	0.245	0.647	0.000	0.000	0.894	0.894
00153120	0.976	0.996	0.410	0.996	0.976	0.996	0.227	0.971	0.579	0.996	0.749	0.996	0.996	0.996
00154184	0.230	0.967	0.168	0.936	0.268	0.218	0.211	0.218	0.151	0.195	0.000	0.971	0.810	0.984
00232213	0.121	0.279	0.399	0.304	0.228	0.276	0.126	0.458	0.243	0.000	0.000	0.000	0.403	0.000
00249450	0.980	0.979	0.980	0.995	0.942	0.995	0.327	0.881	0.651	0.995	0.978	0.980	0.995	0.995
00354176	0.021	0.929	0.221	0.929	0.313	0.930	0.221	0.929	0.281	0.929	0.566	0.929	0.000	0.904
00492401	0.647	0.932	0.231	0.934	0.602	0.847	0.343	0.883	0.292	0.848	0.000	0.910	0.871	0.987
00528181	0.221	0.889	0.472	0.765	0.196	0.889	0.252	0.706	0.164	0.765	0.000	0.732	0.784	0.958
00589302	0.747	0.948	0.950	0.974	0.950	0.948	0.652	0.669	0.950	0.948	0.000	0.824	0.948	0.949
00630878	0.252	0.951	0.000	0.658	0.255	0.747	0.181	0.676	0.265	0.615	0.000	0.835	0.746	0.944
00700032	0.961	0.962	0.608	0.962	0.960	0.962	0.326	0.962	0.614	0.962	0.975	0.963	0.962	0.955
00771600	0.944	0.999	0.618	0.999	0.618	0.621	0.713	0.919	0.000	0.999	0.000	0.985	0.999	0.999
00834538	0.816	0.969	0.319	0.968	0.816	0.968	0.183	0.966	0.816	0.968	0.813	0.968	0.968	0.968
00862084	0.933	0.999	0.157	0.999	0.489	0.905	0.161	0.798	0.124	0.993	0.926	0.975	0.999	0.999
00883892	0.668	0.773	0.590	0.669	0.676	0.789	0.590	0.572	0.590	0.793	0.169	0.000	0.793	0.900
00941601	0.289	0.898	0.332	0.740	0.030	0.873	0.489	0.731	0.338	0.000	0.000	0.916	0.740	0.949
00953658	0.628	0.929	0.622	0.929	0.623	0.929	0.415	0.893	0.415	0.622	0.000	0.844	0.929	0.982
00966049	0.830	0.764	0.836	0.823	0.176	0.735	0.011	0.790	0.000	0.763	0.018	0.000	0.679	0.821